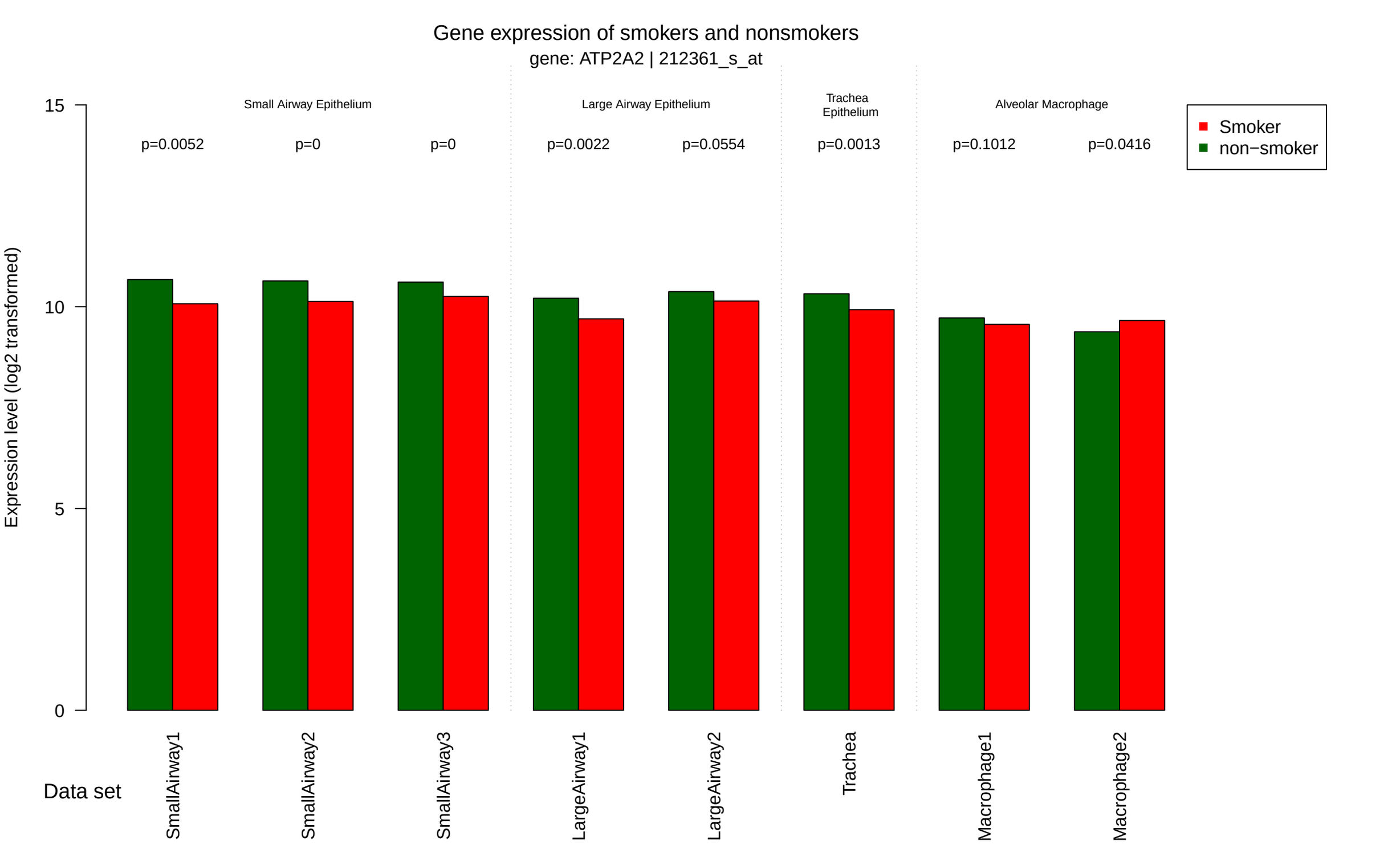
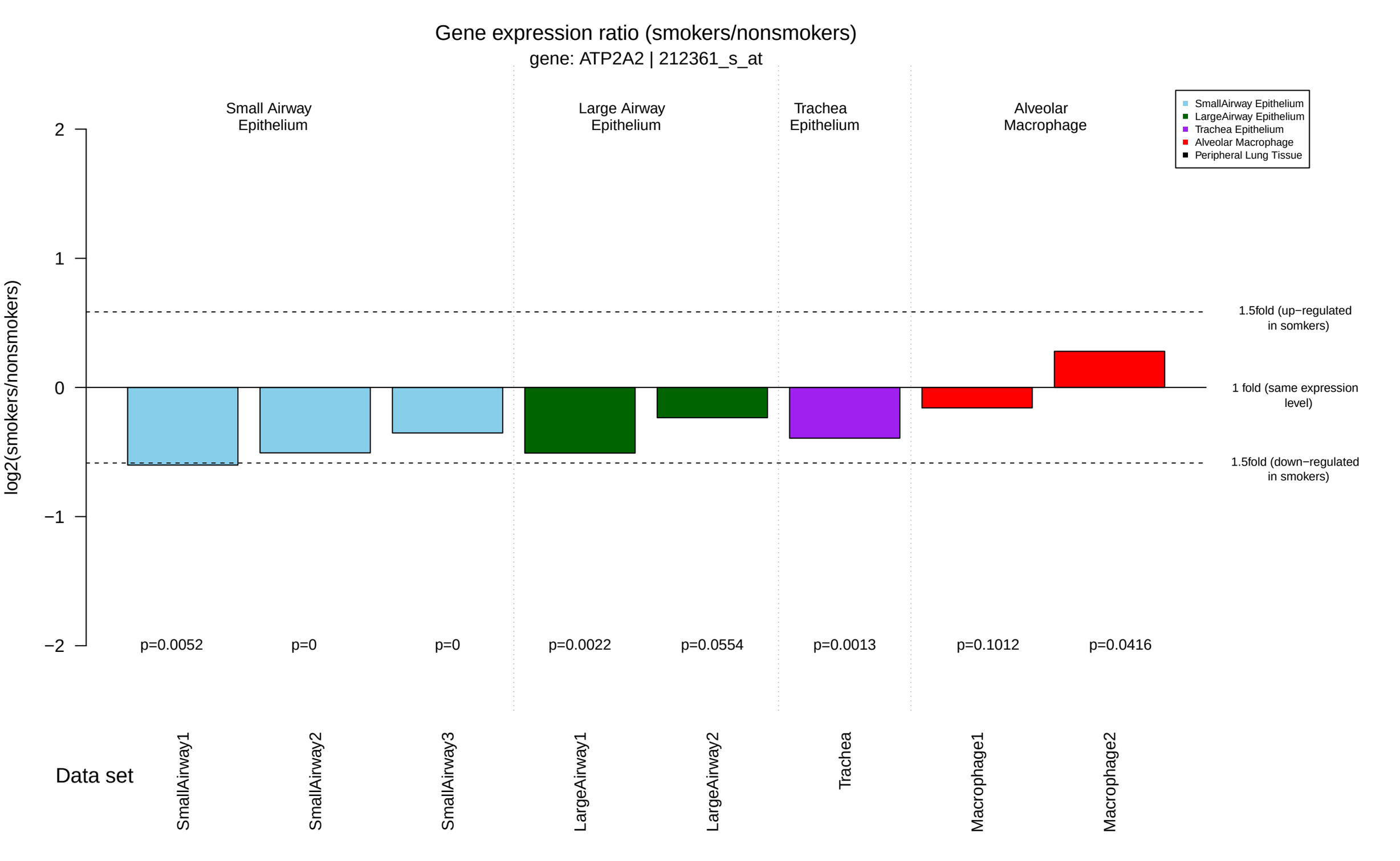
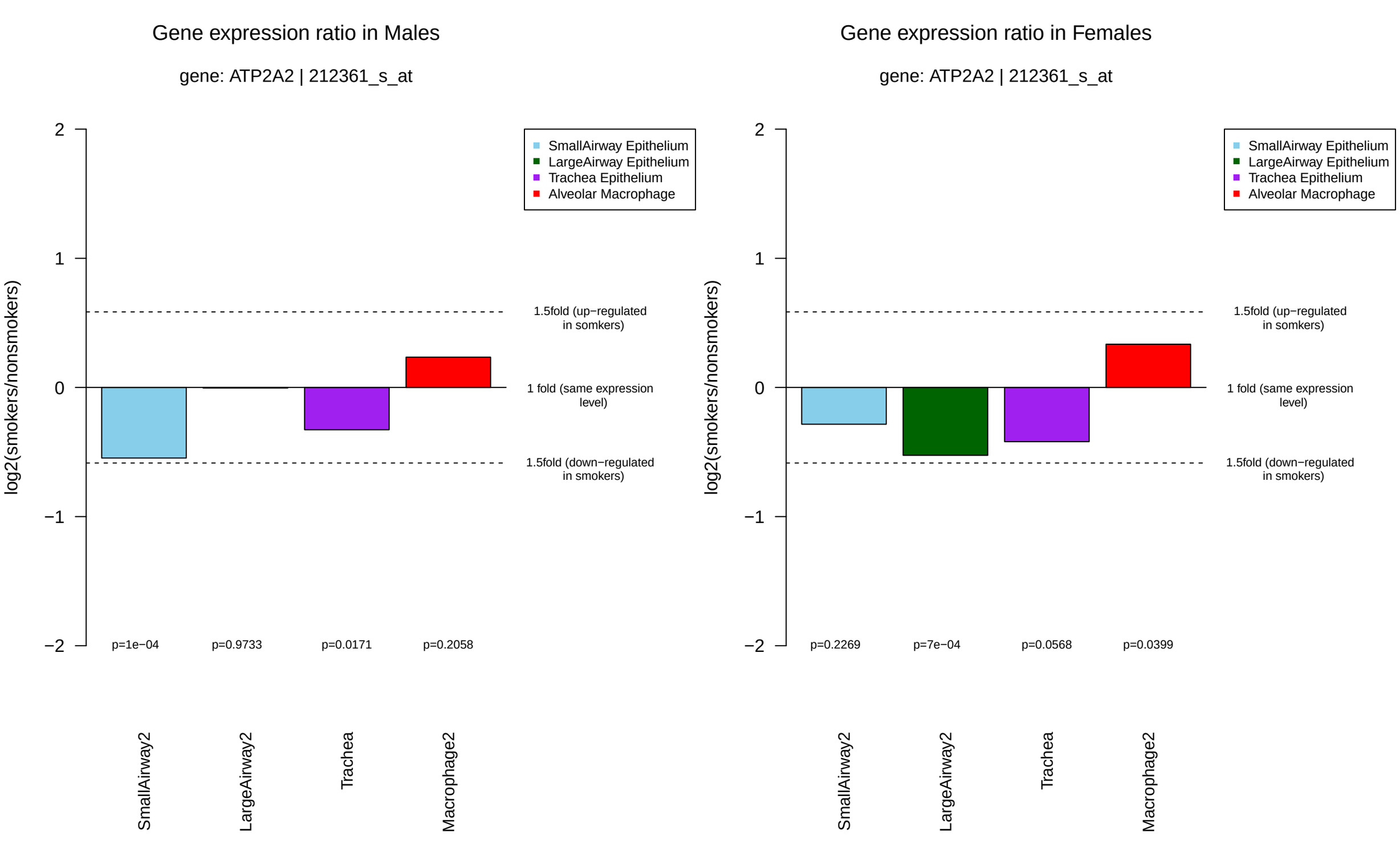
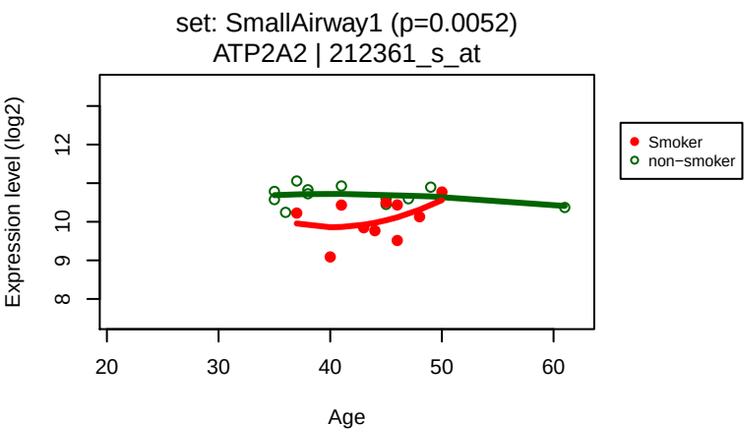
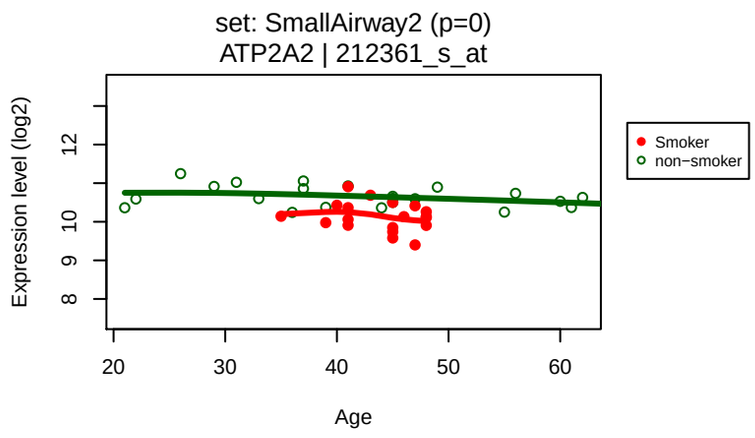
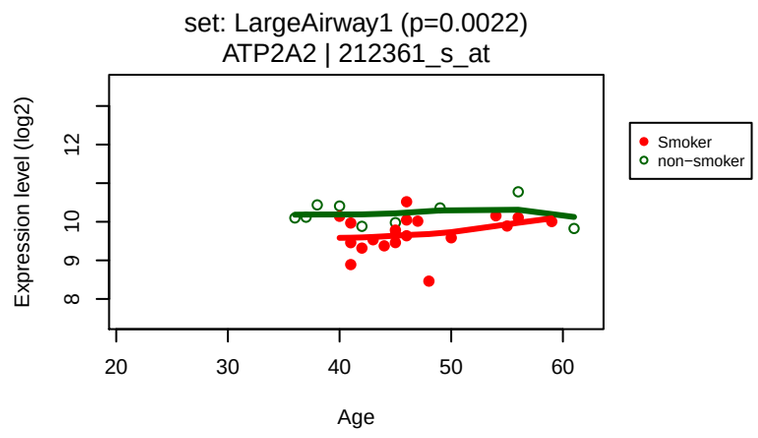

Gene expression of smokers and nonsmokers
gene: ATP2A2 | 212361_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0052
p=0
p=0
p=0.0022
p=0.0554
p=0.0013
p=0.1012
p=0.0416
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

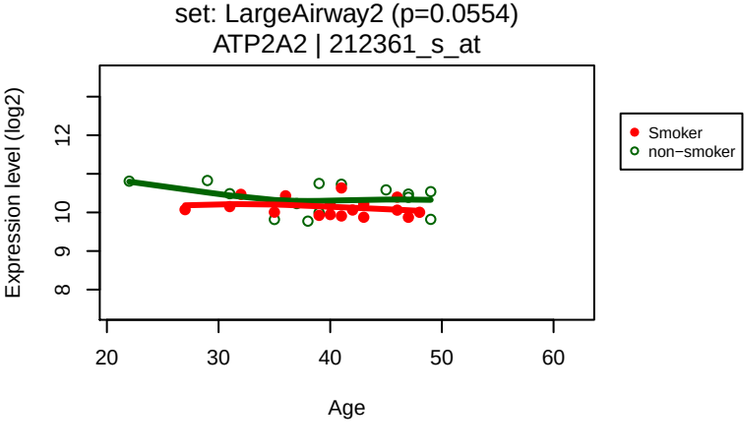

Gene expression ratio (smokers/nonsmokers)
gene: ATP2A2 | 212361_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0052
p=0
p=0
p=0.0022
p=0.0554
p=0.0013
p=0.1012
p=0.0416
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

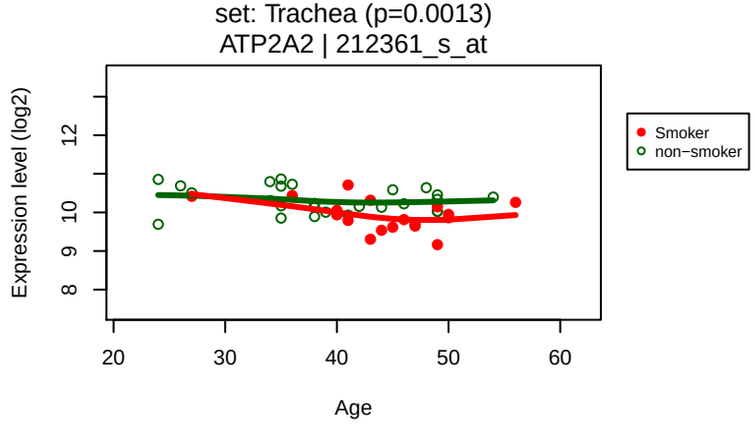

Gene expression ratio in Males
gene: ATP2A2 | 212361_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=1e−04
p=0.9733
p=0.0171
p=0.2058
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP2A2 | 212361_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2269
p=7e−04
p=0.0568
p=0.0399
SmallAirway2
LargeAirway2
Trachea
Macrophage2

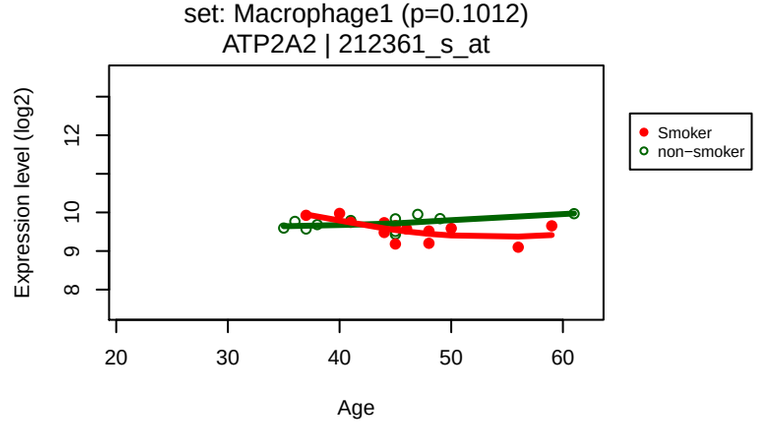

set: SmallAirway1 (p=0.0052)
ATP2A2 | 212361_s_at
Expression level (log2)
8
9
10
12
20
30
40
50
60
Age
Smoker
non−smoker

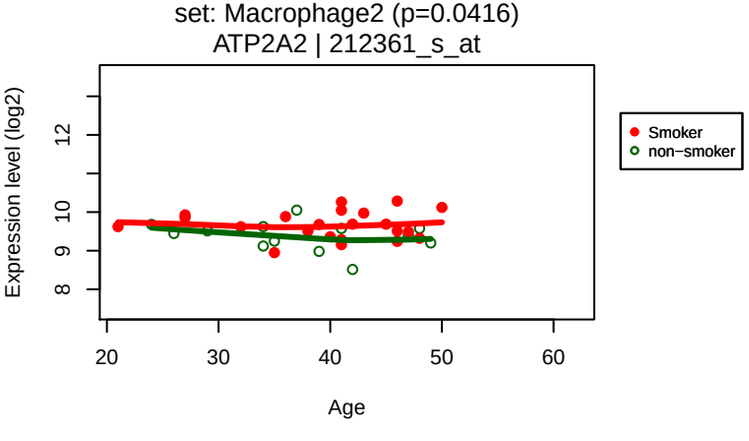

set: SmallAirway2 (p=0)
ATP2A2 | 212361_s_at
Expression level (log2)
8
9
10
12
20
30
40
50
60
Age
Smoker
non−smoker

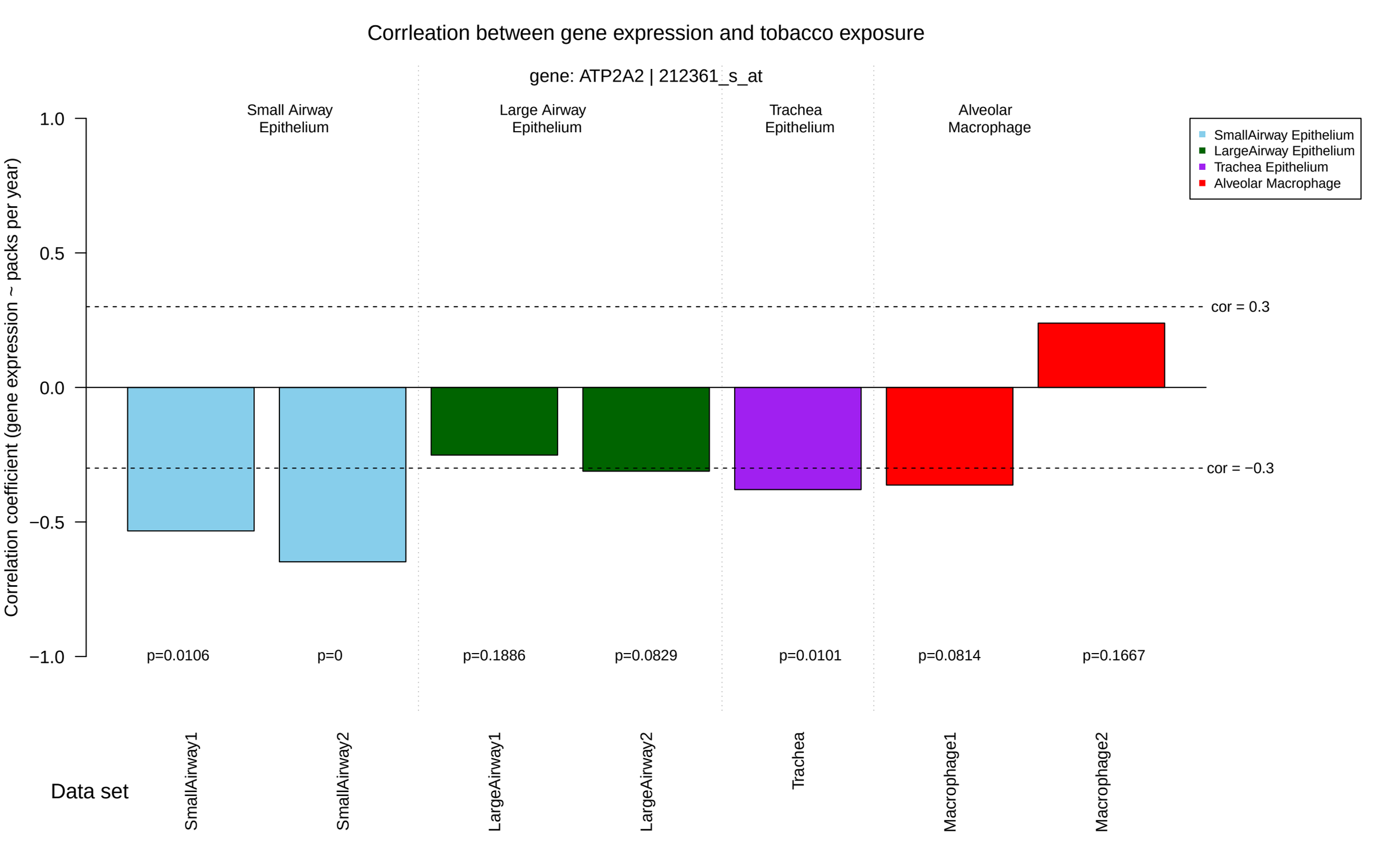

set: LargeAirway1 (p=0.0022)
ATP2A2 | 212361_s_at
Expression level (log2)
8
9
10
12
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0554)
ATP2A2 | 212361_s_at
Expression level (log2)
8
9
10
12
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.0013)
ATP2A2 | 212361_s_at
Expression level (log2)
8
9
10
12
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1012)
ATP2A2 | 212361_s_at
Expression level (log2)
8
9
10
12
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0416)
ATP2A2 | 212361_s_at
Expression level (log2)
8
9
10
12
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP2A2 | 212361_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0106
p=0
p=0.1886
p=0.0829
p=0.0101
p=0.0814
p=0.1667
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2